Q1. A UART receiver is receiving data correctly, but the characters displayed on the terminal are incorrect. List possible causes.

Answer:

UART (Universal Asynchronous Receiver Transmitter) is a serial communication protocol used to exchange data between two devices. For successful UART communication, the transmitter and receiver must have compatible communication settings.

The possible causes of incorrect characters are:

1. **Baud Rate Mismatch:**
   If the transmitter and receiver use different baud rates, the receiver samples the incoming bits at incorrect times, resulting in corrupted characters.

2. **Incorrect Clock/Crystal Frequency:**
   The UART baud rate is generated using the system clock. If the actual crystal frequency differs from the value assumed in the program, the generated baud rate will be incorrect.

3. **Data Bit Mismatch:**
   Both devices must use the same number of data bits. For example, if the transmitter sends 8 data bits but the receiver expects 7 bits, the received characters may be incorrect.

4. **Parity Mismatch:**
   If one device uses parity and the other does not, or if they use different parity settings, communication errors can occur.

5. **Stop Bit Mismatch:**
   The transmitter and receiver should have compatible stop-bit configurations. Incorrect stop-bit settings can result in framing errors.

6. **Incorrect TX and RX Connections:**
   The transmitter output (TX) should be connected to the receiver input (RX), and the grounds of both devices should be connected.

7. **Voltage-Level Mismatch:**
   TTL UART and RS-232 use different electrical voltage levels. A suitable level converter such as MAX232 may be required when interfacing them.

8. **Electrical Noise:**
   Noise or interference in the communication line can change the transmitted bits and cause incorrect characters.

9. **Incorrect UART Register Configuration:**
   Incorrect values in registers such as SCON, TMOD, TH1, or PCON can cause improper UART operation. .

Q2. A UART transmitter is configured for 9600 baud, while the receiver is configured for 4800 baud. What problems may occur?

Answer:

Baud rate represents the rate at which symbols are transmitted through the serial communication channel.

Here:

Transmitter = 9600 baud

Receiver = 4800 baud

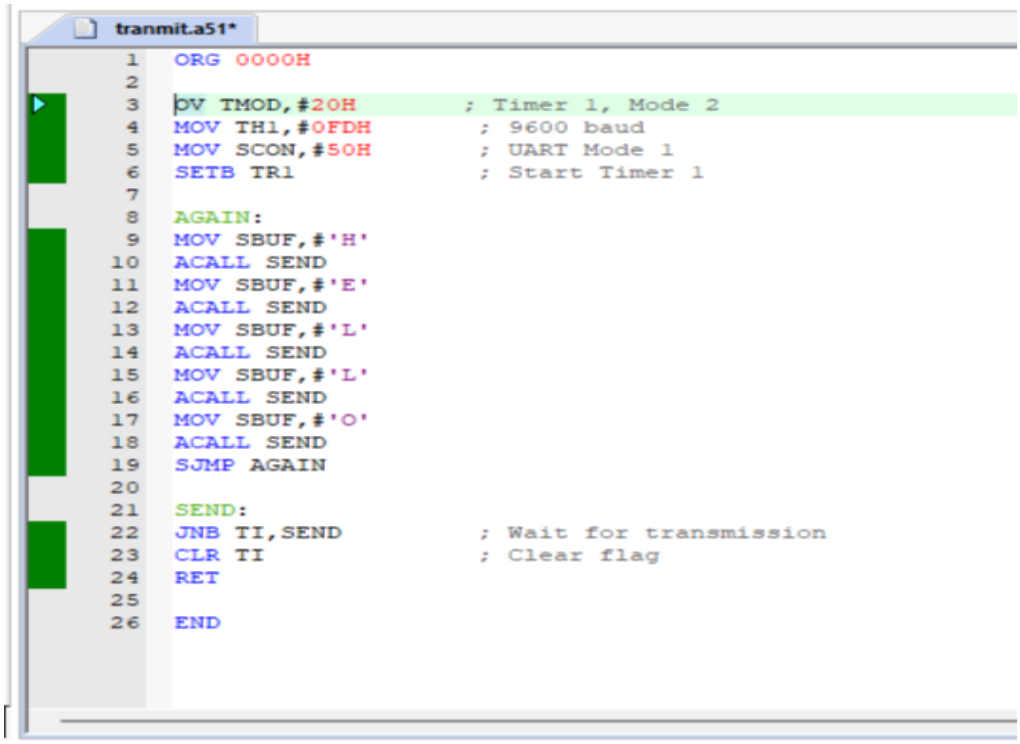
The transmitter is sending data at twice the rate expected by the receiver.

The receiver therefore samples the incoming signal at incorrect time intervals. As a result, the receiver may interpret the transmitted bits incorrectly.

The following problems may occur:

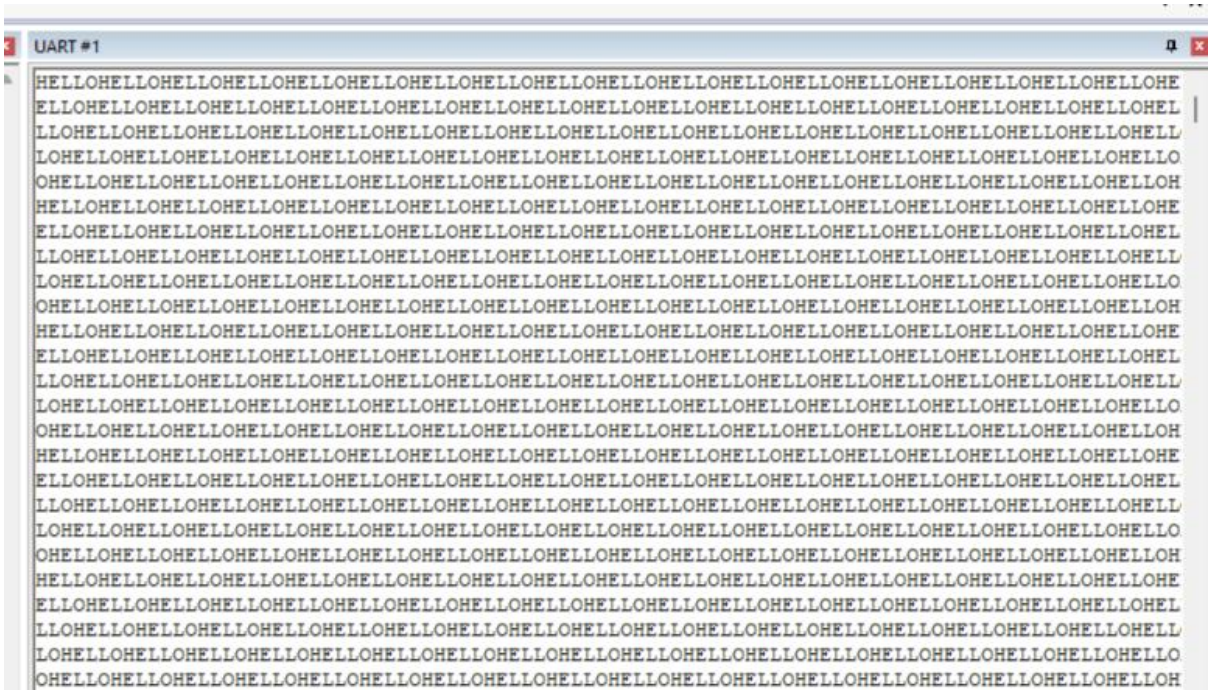
1. **Corrupted characters**
2. **Garbage characters on the terminal**
3. **Framing errors**
4. **Parity errors**, if parity is being used
5. **Loss of data**
6. **Unreliable communication**
7. **Complete communication failure in some cases**

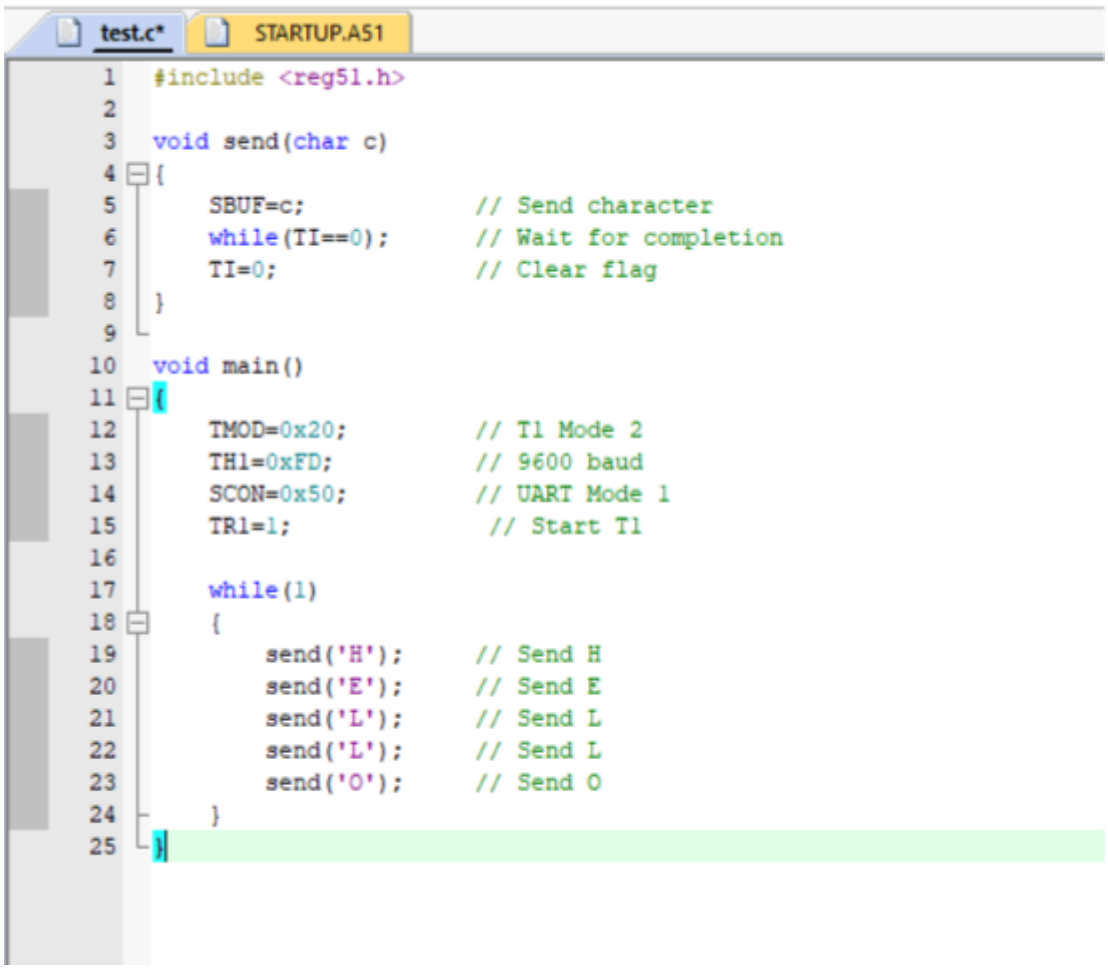
Q3. Write an 8051 Assembly program and Embedded C to transmit the string "HELLO" continuously through UART.

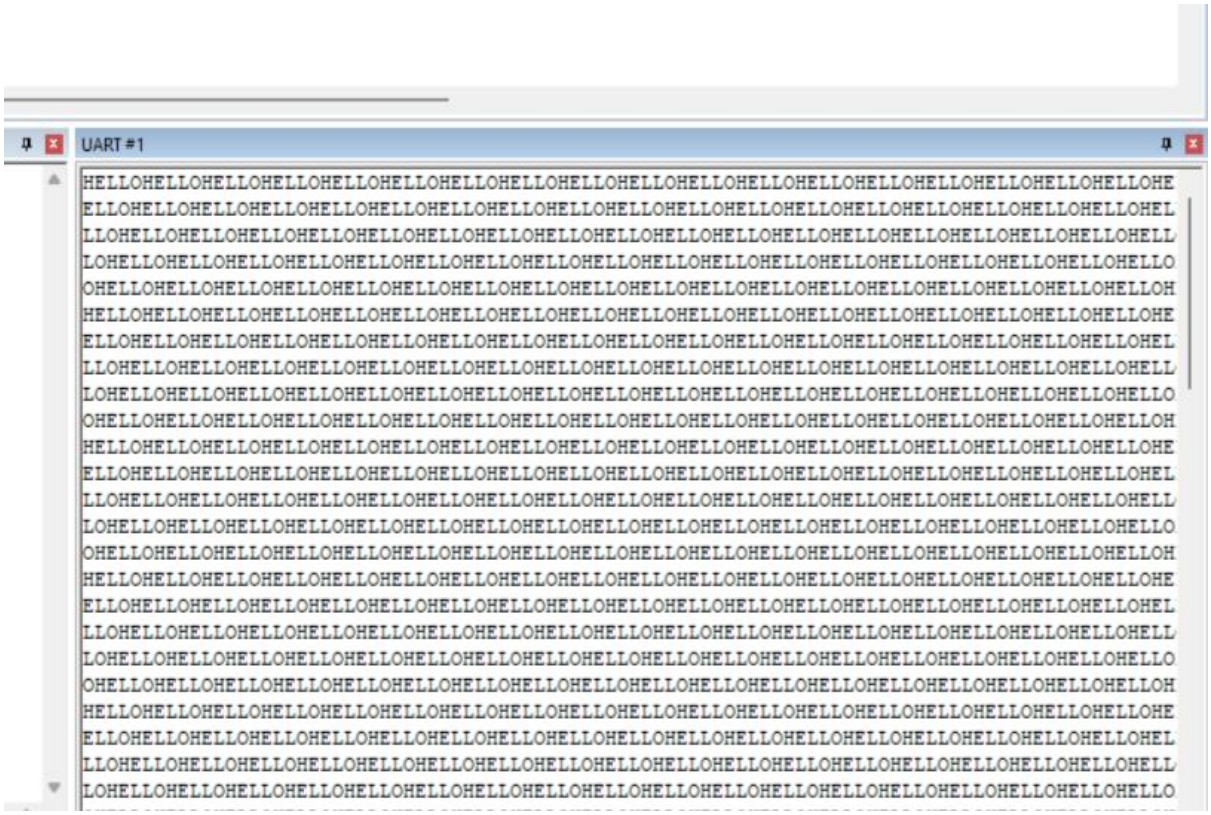


A. Relevant 8051 Registers

Register/Bit	**Value**	**Function and Significance**
TMOD	20H	Configures Timer 1 in Mode 2
TH1	FDH	Timer 1 reload value used for 9600 baud
TL1	FDH	Initial Timer 1 value
SCON	50H	Configures serial port in Mode 1 and enables reception
SBUF	Character value	Serial buffer used to transmit/receive data
TR1	1	Starts Timer 1
TI	1	Indicates that transmission of a character is complete
PCON	00H*	SMOD = 0 in this configuration


UART #1
HELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHE
ELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHEL
LLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELL
LOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLO
OHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOH
HELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHE
ELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHEL
LLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELL
LOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLO
OHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOH
HELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHE
ELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHEL
LLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELL
LOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLO
OHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOH
HELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHE
ELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHEL
LLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELL
LOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLO
OHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOHELLOH


test.c*
STARTUP.A51
1 #include <reg51.h>
2
3 void send(char c)
4 {
5 SBUF=c; // Send character
6 while(TI==0); // Wait for completion
7 TI=0; // Clear flag
8 }
9
10 void main()
11 {
12 TMOD=0x20; // T1 Mode 2
13 TH1=0xFD; // 9600 baud
14 SCON=0x50; // UART Mode 1
15 TR1=1; // Start T1
16
17 while(1)
18 {
19 send('H'); // Send H
20 send('E'); // Send E
21 send('L'); // Send L
22 send('L'); // Send L
23 send('O'); // Send O
24 }
25 }



Q4. A UART transmits data at 115200 baud using the 8N1 format. Calculate the maximum number of bytes transmitted per second, ignoring processing delays.

Answer:

Given:

```
Baud Rate = 115200 bits/second
Format = 8N1
```

Meaning of 8N1:

```
8 = 8 Data Bits
N = No Parity Bit
1 = 1 Stop Bit
```

UART also requires **1 start bit**.

Therefore, each character requires:

```
1 Start Bit
```

```
+ 8 Data Bits
+ 0 Parity Bits
+ 1 Stop Bit
----------------
= 10 Bits
```

Therefore:

$$\text{Bits per character} = 10$$

The baud rate is:

$$115200 \text{ bits/second}$$

Hence:

$$\text{Bytes per second} = \frac{115200}{10}$$

$$\boxed{11520 \text{ bytes/second}}$$

Final Answer:

The maximum number of bytes transmitted per second at 115200 baud using the 8N1 format is **11,520 bytes per second**, ignoring processing delays.